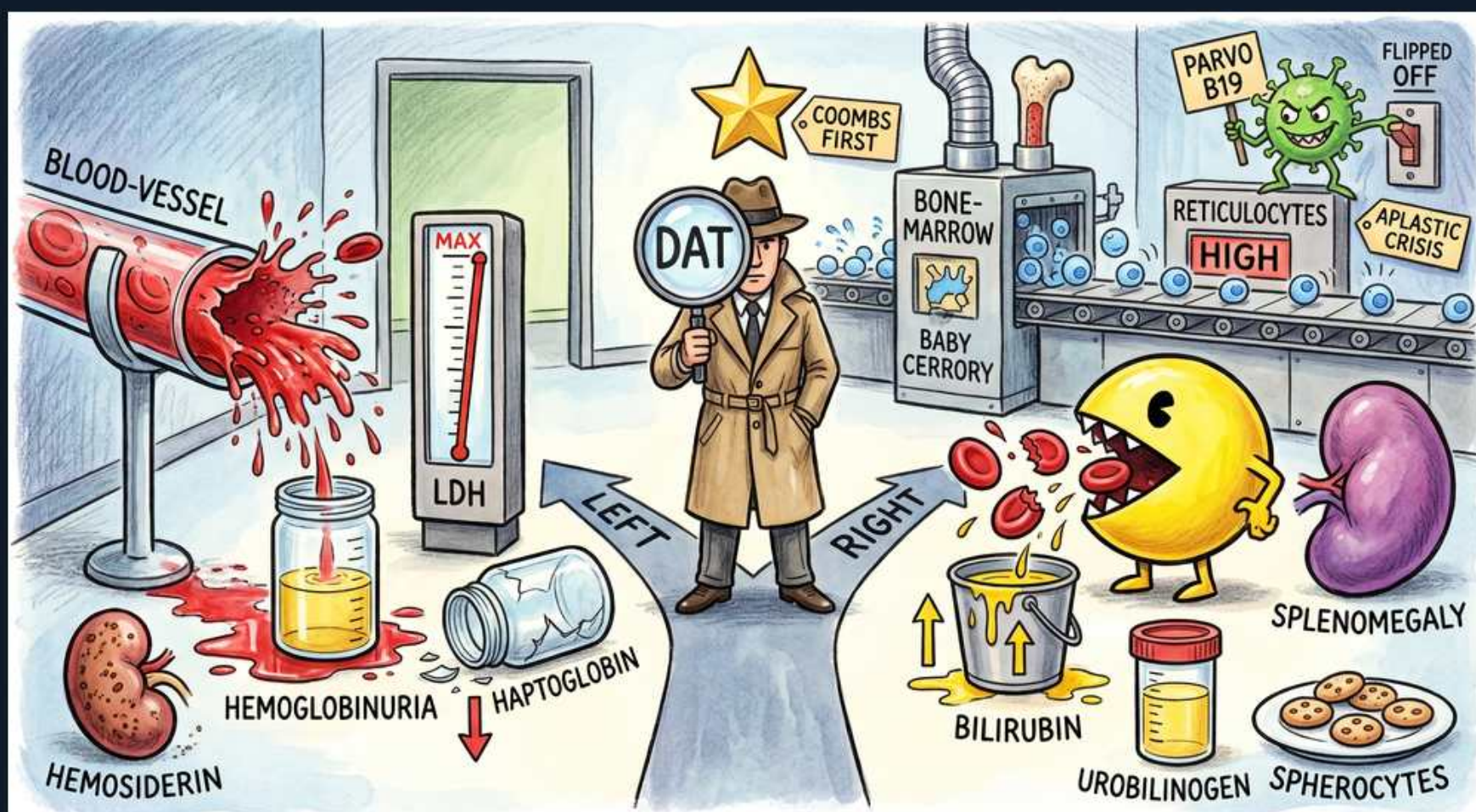


Hemolytic Anemia — Work-up (Intravascular vs Extravascular)



1. DAT — Coombs first

WHAT YOU SEE

The trench-coated detective at center holds up a big round magnifying glass stamped "DAT" — the investigator's first lens through which the whole case is examined.

WHAT IT ENCODES

The Direct Antiglobulin Test (DAT/Coombs) is the first branch point in any hemolysis work-up: positive = immune (autoimmune, drug, alloimmune); negative = non-immune (membrane, enzyme, hemoglobinopathy, MAHA, PNH). Confirm hemolysis itself first with reticulocytes, LDH, haptoglobin and indirect bilirubin.

BOARD TRAP

Ordering an osmotic-fragility or G6PD assay before the DAT — the DAT splits immune from non-immune and reshapes the entire work-up; a positive DAT makes hereditary spherocytosis unlikely (HS is Coombs-negative).

2. Intravascular hemolysis

WHAT YOU SEE

The red blood vessel at upper left has burst open, spraying free crimson blood out into the room — red cells rupturing inside the pipe itself.

WHAT IT ENCODES

Intravascular hemolysis = RBCs lyse inside the vessels, releasing free hemoglobin. Hallmarks: very high LDH, very low/absent haptoglobin, hemoglobinemia → hemoglobinuria → hemosiderinuria. Causes: PNH, MAHA, severe G6PD, transfusion reactions, mechanical/prosthetic valves, march hemoglobinuria.

BOARD TRAP

Labeling splenomegaly + spherocytes as "intravascular" — those are extravascular signs; intravascular shows hemoglobinuria and hemosiderinuria.

3. Hemoglobinuria

WHAT YOU SEE

The clear specimen jar on the left bench filled with red-brown liquid, sitting in a pool of spilled blood — urine stained by free hemoglobin.

WHAT IT ENCODES

Free hemoglobin saturates haptoglobin then filters into urine → red-brown urine that is dipstick-positive for blood but shows NO red cells on microscopy (separating it from true hematuria). Chronic loss causes hemosiderinuria and iron deficiency, classic in PNH.

BOARD TRAP

A positive urine "blood" dipstick with no RBCs read as hematuria — it is hemoglobinuria (or myoglobinuria), a marker of intravascular hemolysis.

4. LDH

WHAT YOU SEE

The tall thermometer labeled "LDH" pegged all the way up to a red "MAX" mark — the enzyme reading off the top of the scale.

WHAT IT ENCODES

LDH pours out of lysed red cells and is markedly elevated in intravascular hemolysis; high LDH with low haptoglobin is the most specific lab combination for hemolysis. It also rises in ineffective erythropoiesis (megaloblastic anemia).

BOARD TRAP

A mildly high LDH alone calling hemolysis — it must be paired with low haptoglobin, high reticulocytes and high indirect bilirubin.

5. Haptoglobin

WHAT YOU SEE

The empty glass bottle tipped over on its side near the center bench with a down-arrow beneath it — the bottle drained and knocked low.

WHAT IT ENCODES

Haptoglobin binds free hemoglobin and is cleared, so it falls (often undetectable) in intravascular hemolysis; low haptoglobin + high LDH is the most specific combo for hemolysis.

BOARD TRAP

Haptoglobin is an acute-phase reactant — inflammation can falsely normalize it, so a normal value does not exclude hemolysis.

6. Hemosiderin

WHAT YOU SEE

The small brown speckled kidney sitting in the bottom-left corner labeled "HEMOSIDERIN" — iron granules dotting the renal tissue.

WHAT IT ENCODES

Chronic intravascular hemolysis deposits iron in shed renal tubular cells → hemosiderinuria (Prussian-blue positive) appearing days after onset. It is a marker of ongoing intravascular hemolysis, classically PNH.

BOARD TRAP

Expecting it immediately — hemosiderinuria appears only days later as iron-laden tubular cells slough.

7. Extravascular (spleen)

WHAT YOU SEE

The large purple bean-shaped spleen on the right side labeled "SPLENOMEGALY" — the organ where tagged red cells are pulled out and destroyed.

WHAT IT ENCODES

Extravascular hemolysis = splenic/hepatic macrophages remove antibody- or abnormality-tagged RBCs. Hallmarks: high indirect (unconjugated) bilirubin, high urobilinogen, splenomegaly, near-normal haptoglobin, spherocytes. Causes: warm AIHA, hereditary spherocytosis, hemoglobinopathies.

BOARD TRAP

Expecting hemoglobinuria — it is NOT a feature of pure extravascular hemolysis.

8. Unconjugated bilirubin

WHAT YOU SEE

The yellow Pac-Man chomping red blood cells beside a bucket of yellow fluid labeled "BILIRUBIN" with up-arrows — heme broken down into golden pigment.

WHAT IT ENCODES

Heme catabolism produces unconjugated (indirect) hyperbilirubinemia with jaundice and pigment gallstones; the conjugated fraction stays normal because the liver conjugates normally.

BOARD TRAP

A high CONJUGATED bilirubin pointing to hemolysis — hemolytic jaundice is unconjugated; conjugated hyperbilirubinemia means hepatobiliary disease.

9. Urobilinogen

WHAT YOU SEE

The specimen cup of yellow urine labeled "UROBILINOGEN" on the right bench — bilirubin breakdown spilling into the urine.

WHAT IT ENCODES

The increased bilirubin load raises urinary and fecal urobilinogen, supporting extravascular hemolysis.

BOARD TRAP

(supportive sign only)

10. Spherocytes

WHAT YOU SEE

The plate of round, dense cookies in the lower right labeled "SPHEROCYTES" — red cells reshaped into uniform little spheres.

WHAT IT ENCODES

Spherocytes on the smear mean part of the membrane was removed by the spleen; they occur in BOTH hereditary spherocytosis (Coombs-negative) and warm AIHA (Coombs-positive). The DAT distinguishes them.

BOARD TRAP

Reading spherocytes as automatically hereditary spherocytosis — warm AIHA is also spherocytic; check the DAT.

11. Splenomegaly

WHAT YOU SEE

The swollen purple spleen on the right, drawn oversized and bulging — the organ enlarged from overwork clearing tagged cells.

WHAT IT ENCODES

Splenic processing of tagged cells enlarges the spleen; common in extravascular hemolysis (HS, warm AIHA, thalassemia).

BOARD TRAP

(supportive sign only)

12. Reticulocytes high

WHAT YOU SEE

The factory conveyor belt of blue cells running along the top right with a sign reading "RETICULOCYTES HIGH" — the marrow line cranking out fresh cells.

WHAT IT ENCODES

The appropriate marrow response to hemolysis is reticulocytosis (corrected retic / RPI > 2–3). A hemolytic anemia with LOW reticulocytes signals a superimposed marrow problem.

BOARD TRAP

A normal/low retic count in acute hemolysis read as reassuring — it actually means the marrow is failing to compensate.

13. Parvovirus B19 — aplastic crisis

WHAT YOU SEE

The spiky green virus glaring beside a "PARVO B19" sign and a wall switch "FLIPPED OFF" leading to the "APLASTIC CRISIS" box — the bug shutting down the marrow line.

WHAT IT ENCODES

Parvovirus B19 infects erythroid progenitors causing transient pure red-cell aplasia. In chronic hemolysis (HS, sickle cell) the shortened RBC lifespan makes marrow shutdown cause a sudden APLASTIC CRISIS: Hb plummets with reticulocytopenia.

BOARD TRAP

A sudden Hb drop with LOW reticulocytes in a known hemolytic patient mistaken for a hemolytic flare — it is aplastic crisis (Parvovirus B19); a flare has HIGH retics.